What is being ported, and what is still not JAX

The configuration is the published ROCKE-3D 2.0 P2SAoM40 setup; the port is a hybrid, and every result says which parts are hybrid.

THE MODEL

+ ModelE2 planet 2.0, rundeck P2SAoM40: Sun-like star, dynamic ocean (13 layers), 40-layer atmosphere on a 72 x 46 grid, 30-minute step.
+ Intended use: reproduce the published climatology (Tsigaridis et al. 2025) at far lower cost so that parameter sweeps of rocky-planet climates become practical.
+ Scored here: one model day (54 steps) from the nov26 restart.

RUNS IN JAX (V100, float64)

+ Atmosphere dynamics, clouds and convection (CONDSE), surface tiles, land (GHY, Ent), ocean, sea ice, lakes.
+ Radiation: SOCRATES ported to JAX (--rad native), or the unmodified Fortran callback (--rad server), always labelled.
+ All behaviour flags default off; the fast configuration adds nine opt-in flags.

STILL HOST OR RECORDED

+ Five step-0 Fortran dump files (about 278 MB) and the nov26 restart; irrigation year fixed at 1850, no fossil groundwater.
+ Day-boundary update, pole columns and Intel libimf math run on the host (libimf is needed: without it cloud thresholds flip in 3-5% of columns).
+ Fast configuration: the cloud-base stage runs in a separate Numba process.

Hybrid boundary: the reference and the JAX inputs come from the same recorded Fortran run, so a start from a bare restart, other seasons and long free runs are not yet demonstrated.

Sources: GOAL (archive); RELEASE_NATIVE_RADIATION_20261012; ACCEPTANCE_CRITERIA s1; STANDALONE_INPUT_AUDIT (archive); SPEED_RUN_RECIPE_20261013

How fidelity is measured

Plain-language wording drafted from the acceptance criteria; the wording has not been reviewed by the owner or a domain expert.

THE IDEA

- The climate model is chaotic: nudging the starting point by one rounding unit gives a slightly different answer later (about 1-3% of the field variability within 5 days in the measured case).
- So the yardstick is the original model's own variation: five Fortran runs started one rounding-unit apart ('members').
- Ratio = how far JAX is from the real Fortran run, divided by the largest member distance from it.

THE RULES (OWNER-SET, ACCEPTANCE s9-10)

- Strict: never beyond 2x at every step and every scored field (8 Oct).
- Relaxed, experiments only: at most 3x from step 3, always shown beside the strict result (11 Oct).
- A ratio of 2.37 means cloud ice is 2.37 times the reference spread away from the reference, not that there is 2.37 times more ice.

TWO DIFFERENT COMPARISONS

- C2 (fidelity): JAX against the real Fortran members. Only C2 supports a statement about ROCKE-3D.
- C1 (port consistency): JAX against our NumPy chain; says nothing about Fortran.

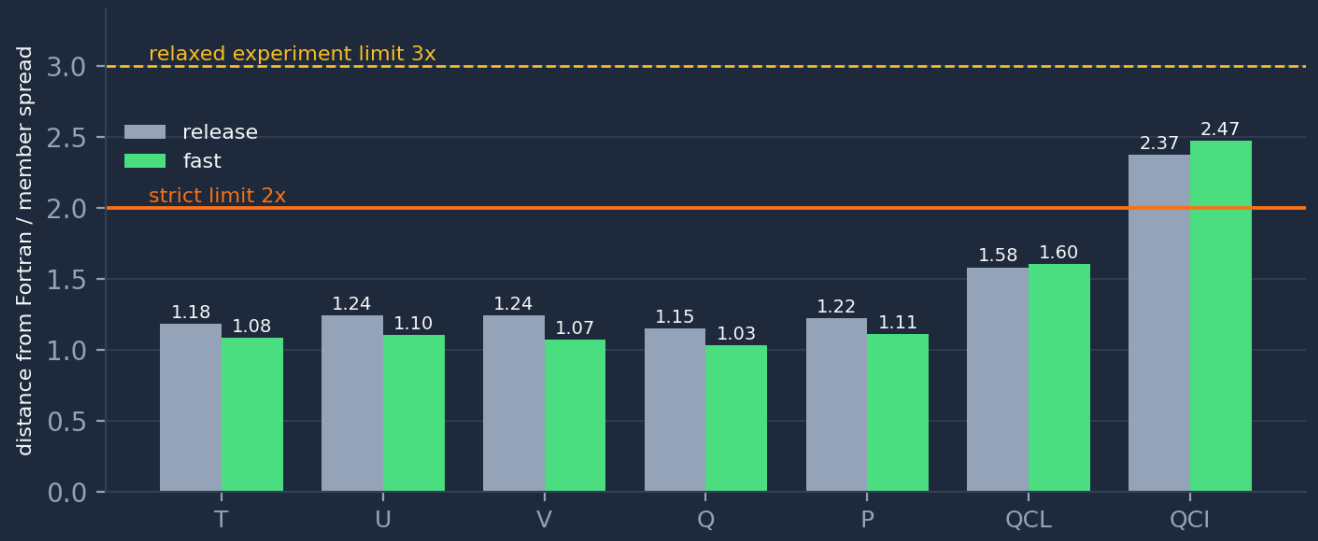
A SEPARATE, SECONDARY CHECK

- Agreement to many decimals (categories A to D) is a precision diagnostic, not the fidelity test; ulp-level differences grow in a chaotic model without meaning physics is missing.

Sources: ACCEPTANCE_CRITERIA s2-4, s9-10; FIDELITY_EXPLAINER_NOTES (archive status_slides, draft); PROJECT_CHRONOLOGY phase B (noise floor D3)

Result: one model day from the nov26 start

Release and fast configurations, native radiation, scored against five one-ulp Fortran members (run of 2026-10-18).



READING THE CHART

- Worst ratio at steps 3 to 53. Only cloud ice (QCI) at step 3 exceeds 2x: 2.37 release, 2.47 fast.
- Steps 0-2 are not shown and are worse: P 20.7 / 24.2 at step 0, QCL 7.9 at step 1, U 3.4 / 2.6 at step 1.
- Day 2: never beyond 2x (worst 1.22 release, 1.11 fast).

Rule	Release	Fast	Source
Strict, all steps (approved)	NOT MET	NOT MET	ACCEPTANCE s9
Strict, steps >= 3 only (secondary)	NOT MET	NOT MET	ACCEPTANCE s9
Relaxed 3x, steps >= 3 (experiment)	MET	MET	ACCEPTANCE s10

Release ratios repeated to two decimals in three runs; fast ratios vary between repeats by up to 0.15 (P) while QCI at step 3 stays 2.47. The fast flags shift QCI by about 0.1 and are not bitwise equal to the release (Ent batching, host cloud stage).

Sources: STATUS_20261016 s5 (runs c, d, e; d207_score.py against members p1-p5); ACCEPTANCE_CRITERIA s9-10; SPEED_STEP_TIME_LOG_20261012 s13 (archive)

Radiation: from a Fortran callback to native JAX

The port started with the original radiation code served from a helper process; a JAX port of the SOCRATES call followed.

STEP 1: THE ORIGINAL CODE AS A SERVER (6-7 OCT)

+ A helper built from the unmodified model runs its radiation step on inputs we supply; 21 of 22 outputs are bitwise equal to the recorded model.
+ Cost 11-14 s per call; every result carried the sentence 'radiation computed by the original Fortran (hybrid component)'.
+ A free-running day through the server kept temperature and winds inside the member spread and global-mean radiative flux within 1 W/m2.

STEP 2: SOCRATES IN JAX (11-12 OCT)

+ Ten consecutive nov26 radiation steps reproduce the recorded server outputs at rounding level (21 fields, AIJ excluded), CPU and V100; one V100 does the whole call in 0.174 s.
+ Day-varying products bitwise against the instrumented server on 48 dates.
+ In the coupled run the median radiation step fell from 10.0 s to 1.9 s.

Limits: only the nov26 window has been run through the full model; the 750 AIJ diagnostics are not ported; the cadence of the native radiation call is unconfirmed; same-code native-radiation repeats are not bitwise and the cause is not localized.

Where the Fortran time went (profile of the unmodified call): aerosol-table interpolation 37%, two-stream flux solvers 26%, optical properties 12%, band driver 8%, math library and I/O 10%, other ModelE side 7%.

Sources: SOCRATES_SPIKE_RESULT s2; ACCEPTANCE_CRITERIA s12a-12d; RELEASE_NATIVE_RADIATION_20261012; PROJECT_CHRONOLOGY D152-D157 (archive)

What this evidence cannot say

Each statement has a scope; a passing score does not imply broader validity.

ABOUT THE TEST

- It does not prove the physics is complete. Whether an explicit physics-coverage check exists has not been established.
- Trajectories are never compared; only distributions against the member spread.
- One start date (nov26), one scored day, V100 float64.
- The reference and the JAX inputs come from one recorded Fortran run.

ABOUT REPEATABILITY AND LONG RUNS

- Native-radiation repeats of the same code differ: raw surface radiative flux by up to 543 W/m2 pairwise in 54-step runs; zenith bitwise equal; cause not localized.
- GPU exp/pow differ from CPU libm by ulps; the owner-approved route is an Intel libimf host callback for the headline comparison.
- The 30-day shadow run is invalid after step 1288: a Fortran branch (qdynam_do_z_extra) that the port does not execute.
- The fast flags were run for 108 steps only, and they change arithmetic (Ent batching is not bitwise equal; QCI at step 3 moves 2.37 -> 2.47).

Describe this as a well-documented experimental result, not as a validated or production-ready port.

Sources: REPEATABILITY_AND_FLAGS_REGRESSION_20261014; STATUS_20261016 s4; ACCEPTANCE_CRITERIA s8, s10; SPEED_STEP_TIME_LOG_20261012 s13-14 (archive)

Open science questions and the check for each

Ordered by how much they would change the conclusions.

Question	Proposed check	State
Why does cloud ice exceed 2x at step 3 (2.37, 2.47)?	CPU day with Intel libimf (cpu_day_run_imf.py). Caution: the fast run already uses libimf for the cloud-base stage and still gives 2.47.	dry-run only
Why are same-code native-radiation repeats not bitwise?	Localize the staged-input path or flag responsible; a flags-off short window repeated bitwise.	cause not found
Is the physics complete?	An explicit physics-coverage check, separate from the spread test.	not established
Other seasons and longer runs?	More start dates (restart, step-0 dumps and members exist for nov26 only); fix the 30-day failure.	not done
Does host NumPy on AVX-512 nodes move the ratios?	Repeat a day with the AVX-512 paths disabled (they change 65 bitwise tests).	untested
Which published target?	100-year annual mean or a shorter run; to be confirmed with the project lead.	open

Sources: STATUS_20261016 s4; QA_HANDOFF_20261016 and GOAL (archive); fullfidelity/qa_20261016/ENVIRONMENT.md; DATA_AND_PATHS